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Short communication

Improvement of methane generation capacity by aerobic pre-treatment of organic waste with a cellulolytic *Trichoderma viride* culture

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ABSTRACT

Trichoderma viride is known as a potent cellulose decomposer and was successfully used to improve and accelerate the decomposition process of aerobic composting. In contrast, the role of fungi as pre-treatment organisms for anaerobic digestion is not clear, since the fast aerobic decomposition is thought to be responsible for a rapid depletion of easily available nutrients, leading to a lack of these for the anaerobic community. In the present study carried out in lab-scale, the application of *T. viride* for the aerobic pre-incubation of organic matter derived from the inlet port of a 750,000 L anaerobic digester led to an increase in total gas and methane production in a subsequent anaerobic digestion step. A high cellulase activity caused by the addition of *T. viride* seemed to be responsible for a better nutrient availability for anaerobic microorganisms. Therefore, aerobic pre-incubation of organic residues with *T. viride* for subsequent anaerobic digestion is a promising approach in order to increase methane yields.
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1. Introduction

Lignocellulose represents a relatively unused source for biogas (Hendriks and Zeeman, 2009) and consists of three types of polymers – cellulose, hemicellulose and lignin. It represents the major component of biomass, comprising around half of plant matter produced by photosynthesis (Perez et al., 2002; Sanchez, 2009). Many microorganisms are capable of degrading cellulose and using its breakdown products as a carbon source. Fast degradation of cellulose mainly takes place in aerobic environments by various groups of fungi through the production of extracellular cellulolytic enzymes. In anoxic environments cellulose decomposition is possible but occurs slowly, requires mixed populations of different bacteria (Leschine, 1995), and often takes place within cellulosomes (Adney et al., 1991; Desvaux, 2005; Lynd et al., 2002).

Trichoderma viride (Ascomycota, Euascomycetes, Hypocreales, Hypocreaceae) (de Hoog and Guarro, 1995) is an ubiquitous saprophytic fungi (Samuels, 1996) widely found in soils and known to produce cellulases (Lieckfeldt et al., 1999), chitinolytic- (Lorito, 1998), and hemicellulolytic enzymes (Biely and Tenkanen, 1998). *T. viride* was successfully used to improve and accelerate the decomposition of an aerobic composting process (Illmer et al., 2007). In matrices such as compost, interactions with other organisms can lead to complete degradation of cellulose (Sanchez, 2009). In contrast, for use as pre-treatment organisms for subsequent anaerobic digestion the role of any aerobic fungi is not clear, since the fast aerobic decomposition might be responsible for a rapid depletion of easily available nutrients, leading to a lack of these for a subsequent anaerobic community.

In biogas plants processing waste with high cellulose content residues, the initial hydrolysis phase is often the process rate-limiting step, thus hindering the subsequent microbial populations producing the desired biogas(es) (Lynd et al., 2002). Different approaches have been made to enhance both turnover rates in the hydrolysis phase as well as the extent of hydrolysis in the anaerobic digestion process by a pre-treatment step. These include thermal treatment (e.g. Appels et al., 2010), microwave technology (e.g. Eskicioglu et al., 2007), the direct application of enzymes (e.g. Gerhardt et al., 2007; Mendes et al., 2006), alkaline pre-treatment (e.g. Chiu et al., 1997), combined treatments (e.g. Bougrier et al., 2006) and fungal pre-treatment of fermentation substrates with selected strains of *Sporotrichum*, *Aspergillus*, *Fusarium*, and *Penicillium* (e.g. for orange processing waste (Srilatha et al., 1995)). As far as we know *T. viride* has not yet been used as a pre-treatment organism for the improvement of the methane potential from organic waste designated for anaerobic digestion.

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The aim of the present study was to show the benefit of direct application of *T. viride* in an aerobic pre-treatment step to enhance subsequent anaerobic methane production. Measuring cellulase activity was used to demonstrate the potential of *T. viride* to degrade cellulose in organic residue, thus making nutrients available for the subsequent anaerobic population resulting in an increased methane production rate.

2. Material and methods

2.1. Cultivation of T. viride

T. viride (Centraalbureau voor Schimmelcultures CBS 456.96) was cultivated on malt extract agar plates (malt extract 30 g, peptone 5 g, agar 20 g, 1000 ml distilled water, pH 6.5) at 25 °C. For liquid cultures, a malt extract solution (malt extract 30 g, peptone 5 g, 1000 ml distilled water, pH 6.5) was used, inoculated with a *T. viride* from agar plates and cultivated at 25 °C at 120 rpm for 3 days. Mycelia were harvested by centrifugation with 5000× g and washed with 0.9% NaCl solution.

2.2. Aerobic pre-treatment

Bio-waste (Table 1) usually treated in a 900 m^3 anaerobic digestion plant in Roppen, Austria, was used for the aerobic pre-incubation step. This waste was used to investigate conventional sludge similar to that often treated in large-scale digestion plants. An aerobic pre-treatment step was selected since the waste is usually stored under aerobic conditions for a few days (depending on the amount of delivered organic waste) without additional treatment. For a description of running parameters of the digester and the composition of the sludge refer to Illmer et al. (2009) and Wagner et al. (2009). Some basic parameters are given in Table 2. Prior to use, the organic waste was diluted 1:5 with distilled water. For the aerobic pre-treatment step, 100 g of the diluted waste was weighted into 250 ml Schott flasks and inoculated with 5.0 g (variant 5 g) or 15.0 g (variant 15 g) of *T. viride* fresh weight with a dry matter content of 5.14% (0.257 g dry matter for the 5 g variant and 0.771 g dry matter for the 15 g variant). Beside the cellulolytic potential of *T. viride*, this organism was selected because of its wide pH growth range between 1.5 and 9 (de Hoog and Guarro, 1995). As a control (variant C), diluted waste was used without additional components. All variants were done in 3 replicates. The flasks were closed with cotton stoppers and incubated at 25 °C for 4 days.

2.3. Anaerobic treatment

After the 4 days of aerobic pre-treatment the samples were mixed and 10.0 g (±0.10 g) were weighted into serum bottles and flushed with gas consisting of 70% N_2 and 30% CO_2 for 2 min to remove headspace oxygen. The bottles were closed with butyl rubber septa and subsequently inoculated with 5 g of 1:5 diluted digester outlet sludge (inoculum) to start the anaerobic decomposition process. Additionally 5 g of the anaerobic inoculum were tested toward methane production from the pure inoculum. All variants were carried out in triplicate. The flasks were incubated at 52 °C for 18 days.

Table 1
Selected parameters of the bio-waste used for anaerobic digestion in the biogas reactor (Roppen, Austria) reflecting raw waste.

Parameter	Means (±SD)
pH	5.11 (0.156)
Dry weight	38.5% (0.02)
Organic matter	69.9% (0.03)
NH_4^+–N	0.69 g kg^{-1} (0.260)
Total carbon	35.40% (3.424)
Total nitrogen	2.15% (0.229)
C/N ratio	16.50 (1.249)

Table 2
Selected parameters of the digestion plant and the organic waste derived from the anaerobic digestion plant's input sampling port used in the experiments. Data reflecting raw waste mixed with process water at the beginning of the digestion process. AO: acridine orange.

Parameter	Means (±SD)
Type of digester	Anaerobic continuous solid phase digester
Waste	Organic fraction of household waste (bio-waste) (Table 1)
Retention time	29.4 d (4.5)
Temperature	53 °C (0.3)
Dry weight	26.2% (2.0)
pH	7.9 (0.44)
Organic matter	61.8% (3.55)
NH_4^+–N	3.2 g kg^{-1} (0.46)
Acetic acid	1.9 g kg^{-1} (1.52)
Propionic acid	1.3 g kg^{-1} (1.15)
Total cell counts (AO-staining)	1.4 * 10^{11} cells ml^{-1} FS (1.07 * 10^{11})

Modified after Wagner et al., 2009.

2.4. Experimental design

For the aerobic pre-treatment *T. viride* was added to diluted waste and together with not treated controls incubated for 4 days. Samples for the determination of cellulase activity, organic acids, glucose concentration, and total carbon and nitrogen were taken at days 0, 1, and 4. Subsequently, the aerobically treated samples (with and without *T. viride* application) were used for the anaerobic incubation and inoculated with digester outlet sludge (inoculum) (anaerobic incubation start = day 0). After inoculation, the flasks were brought to temperature, the headspace gas concentrations of H_2, CO_2 and CH_4 were determined by GC measurement, and the overpressure resulting from the increase in temperature was removed by a cannula. Overpressure and gas concentration of H_2, CO_2, and CH_4 in the flasks were measured at days 1, 4, 11, and 18. Samples for organic acids and glucose determination were taken at days 0, 1, 4, 11, and 18. The amount of gas production was calculated from overpressure in the headspace and gas composition.

2.5. Analytical methods

Overpressure in flasks was determined by a pressure measuring instrument (Greisinger, Germany). Analysis of headspace gas concentration of H_2, CO_2, and CH_4 was conducted by GC-analysis using a Shimadzu GC2010 equipped with a Restek Shin Carbon TS 100/120 mesh column (length 2 m, 1 mm inner diameter), a wide bore injector and a thermal conductivity detector as described in Wagner et al. (2012). Injection volume was 15 µl. For the analysis of short chain fatty acids (SCFA) via HPLC, a Shimadzu Prominence System was used with a BioRad AMINEX HPX-87H column, an isocratic system with 5 mM H_2SO_4, 65 °C, and a flow rate of 0.7 ml min^{-1}. Detection was conducted via a UV-detection system at 210 nm using an external standard (Sigma–Aldrich, Germany). Sample preparation using filtration and centrifugation was carried out as described previously (Wagner et al., 2010). For glucose analysis on a Shimadzu Prominence System (HPLC), an aliquot of the samples used for SCFA analysis was injected (25 µl) onto a Phenomenex Fast Fruit Column (Phenomenex, Germany) running with distilled water. For detection, a light scattering detector (LSD) (Shimadzu) was used, and the calibration was performed by an

external standard. Total carbon and total nitrogen content in the liquid fraction (dissolved) was determined using a TruSpec® CHN analyser (Leco, Germany) according to the manufacturer's protocols. Cellulase activity was determined following the instructions of Schinner et al. (1996).

2.6. Statistical analysis

Statistical analysis was performed using the Software package Statistica 8.0 (StatSoft Inc., USA).

3. Results and discussion

3.1. Aerobic pre-treatment

Within the 4 days aerobic incubation period transient rates of cellulase activity could be observed. A significant difference ($p < 0.05$) was found after 4 days of aerobic incubation between the variants, indicating a high activity of the added *T. viride* mycelia. The highest cellulase activity could be attributed to the 15 g variant, followed by – and significantly different from – the 5 g variant. The lowest cellulase activity (significant $p < 0.05$) was observed for the control samples without *T. viride* inoculation/pre-treatment.

Differences could also be observed in the concentration of organic acids. The 15 g variant showed distinct and highly significant ($p < 0.01$) differences compared to the other treatments. Beside others, butyric and, most notably, acetic acid concentrations in the 15 g variants were dramatically increased, resulting in a higher amount of total organic acid concentration (formic-, acetic-, propionic-, iso-butyric, and butyric acid). The increased acetic acid concentration (Fig. 1) in particular might have been beneficial for the subsequent anaerobic treatment, since this organic acid represents the most important precursor for methane in such environments. As a consequence, the high acetate concentration in the 15 g variant also caused an increased acetic:propionic acid ratio (Ac/Pr ratio) (<10) (data not shown), which is usually thought to be advantageous for biogas plants. Generally, an Ac/Pr ratio above 2 is considered as a minimum requirement for stable reactor performance, although distinctly lower ratios were also shown to occur without causing problems as demonstrated by Pullammanappallil et al. (2001) and Illmer et al. (2009).

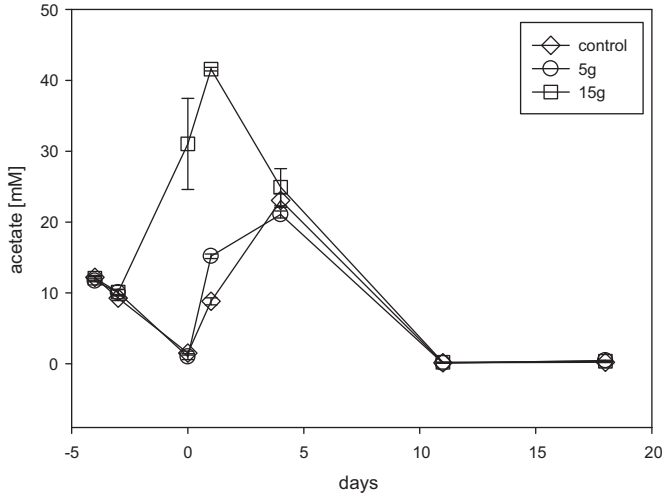


Fig. 1. Acetate concentration (mean ± SD) during the aerobic pre-incubation (day −4 to 0: aerobic pre-treatment; day 0–18: anaerobic digestion). C: control without *T. viride* inoculation, 5 g: 100 sludge inoculated with 5 g (fresh weight) *T. viride* mycelia; 15 g: 100 sludge inoculated with 15 g (fresh weight) *T. viride* mycelia.

Table 3
Mean (±SD) of total gas and total methane production [ml] calculated from overpressure in the flasks after 18 days of anaerobic incubation. C: control without *T. viride* inoculation, 5 g: 100 sludge inoculated with 5 g (fresh weight) *T. viride* mycelia; 15 g: 100 sludge inoculated with 15 g (fresh weight) *T. viride* mycelia. Different bold characters indicate significant differences within a column.

	Total gas [ml]	Total methane [ml]
Control	66.16 (8.915) (**a**)	11.41 (3.100) (**a**)
5 g	100.79 (16.564) (**b**)	23.42 (6.586) (**b**)
15 g	150.19 (7.231) (**c**)	47.36 (2.397) (**c**)

For total carbon and total nitrogen, no differences could be observed during the aerobic incubation step.

3.2. Anaerobic incubation

After 18 days of anaerobic incubation the total gas yield was significantly increased resulting from the aerobic pre-treatment with *T. viride* compared to the control without pre-treatment (Table 3). Most importantly, it was also possible to significantly enhance the methane yield for both *T. viride* treatments (5 g and 15 g; Table 3, Fig. 2). From all treatments the highest methane

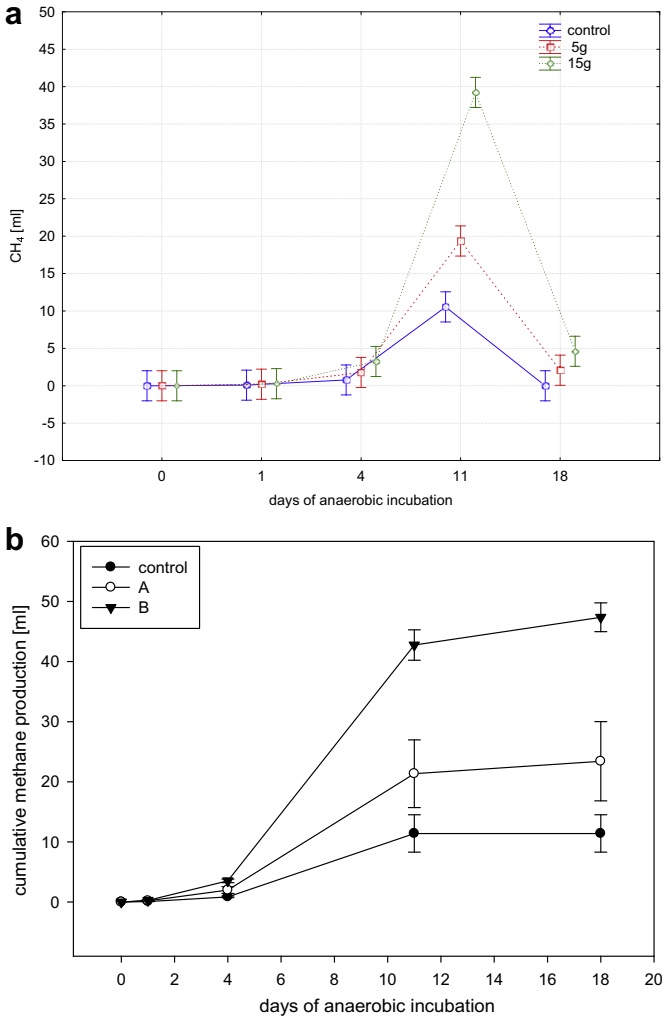


Fig. 2. Methane generation within 18 days of anaerobic incubation; a: methane production from different treatments [ml] (interaction plot from ANOVA); b: cumulative methane production (±SD). C: control without *T. viride* inoculation, 5 g: 100 sludge inoculated with 5 g (fresh weight) *T. viride* mycelia; 15 g: 100 sludge inoculated with 15 g (fresh weight) *T. viride* mycelia.

production rates were observed at day 11 of anaerobic incubation, with the 15 g variant presenting the highest rate. After day 11, the methane production rate decreased again for all treatments, although the pre-treated variants still showed a methane production (Fig. 2a), whereas the control did not. In comparison to other treatment methods this yield reflects a high methane-forming efficiency. A combined alkaline and ultrasonic treatment (although in a different matrix and with a different digestion system) led, for example, to an increase in the methane yield of 150.4 percent (Kim et al., 2010), whereas an over 400 percent increase was detected in the present investigation.

The total amount of organic acids, and especially acetic acid, was significantly higher for the 15 g treatment in the first days of anaerobic incubation compared to the control samples, and decreased after the major part of total methane had been produced (Fig. 2a). Therefore, a high availability of organic acids seems to be one of the most important factors for the successful production of methane. By the application of *T. viride* for an aerobic pre-incubation step this availability was increased significantly. Since acetate is the most important precursor of methane, this might be responsible for the subsequent increase in methane production.

4. Conclusions

The application of *T. viride* for the aerobic pre-incubation of organic matter led to a significant and drastic increase in total gas and methane production in a subsequent anaerobic incubation step. The measured high cellulase activity caused by the addition of *T. viride* seems to be responsible for a better nutrient availability for anaerobic microorganisms. It could be shown that an aerobic pre-treatment with *T. viride* is not responsible for a fast depletion of nutrients, but instead results in an increase in the available volatile fatty acid concentration for the subsequent anaerobic microbial population. Therefore, lab results suggest a high potential for a large-scale applicability of the proposed pre-treatment method using *T. viride*.

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